



3D Urban Building Energy Assessment for BIPV Applications using Geospatial Techniques

Report submitted for completion of training

Programme of Space Applications Centre

Submitted by

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Ahmedabad, Gujarat

13 October 2025 to 14 February 2026

CERTIFICATE

This is to certify that **Mr. Vaibhav Dineshbhai Nai**, a student of Bachelor of Science in Information Technology, has successfully completed his **Research Internship** at **Indian Space Research Organisation (ISRO) – Space Applications Centre (SAC), Ahmedabad (Bopal Campus)** during the period from 11 October 2025 to 14 February 2026.

During the internship, he carried out his project titled **“3D Urban Building Energy Assessment for BIPV Applications using Geospatial Techniques”** under the guidance of **SMT. SHWETA MISHRA, SCIENTIST - 'SF', Space Applications Centre (SAC), ISRO.**

As part of his work, he was involved in geospatial data processing, 3D urban model Development, shadow and solar potential analysis, and visualization of building-integrated photovoltaic (BIPV) outputs. He demonstrated strong analytical & statistical ability, programming proficiency, and a research-oriented approach throughout the internship period.

His conduct, dedication, and overall performance during the training were found to be _____. We wish him every success in his future academic endeavours and research career.

Signature -

Guide's Name -

Designation -

Organization -

DECLARATION

I hereby declare that the Internship Report entitled “**3D Urban Building Energy Assessment for BIPV Applications using Geospatial Techniques**” is an original and authentic record of the work carried out by me at **Indian Space Research Organisation (ISRO) Space Applications Centre (SAC), Ahmedabad (Bopal Campus)** during the period from **11 October 2025 to 14 February 2026**, under the guidance and supervision of **Smt. Shweta Mishra, Scientist ‘SF’, Space Applications Centre (SAC), ISRO.**

The work presented in this report is the outcome of my own efforts and learning during the internship period. The data, methodologies, observations, and results discussed herein are based on the practical tasks and research activities carried out during the training. Wherever reference has been made to the work of others, appropriate acknowledgements and citations have been provided.

I further declare that this report has not been submitted, either wholly or in part, to any other university, institution, or organization for the award of any degree, diploma, certification, or academic credit. The content of this report has been prepared in accordance with the academic guidelines and in compliance with the confidentiality norms and ethical standards of **Indian Space Research Organisation (ISRO) – Space Applications Centre (SAC), Ahmedabad.**

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I am deeply thankful to **SMT. SHWETA MISHRA, SCIENTIST - 'SF'**, Space Applications Centre (SAC), ISRO, for her continuous guidance, encouragement, and technical support throughout the internship period. Her valuable insights and constructive suggestions played a crucial role in the successful completion of this work.

I would also like to acknowledge all the scientists, engineers, and staff members of **Space Applications Centre (SAC)** for their cooperation and support during the internship. The learning experience and professional environment at SAC have greatly contributed to enhancing my technical knowledge and research aptitude.

I express my sincere thanks to **Monark University** for granting permission to undergo this internship and for providing the necessary academic support. I am also grateful to my teachers and mentors for their encouragement and guidance throughout the course of my study.

Finally, I would like to thank my family and friends for their constant motivation and support during the internship period.

ABSTRACT

Rapid urbanization has led to an increased demand for sustainable and renewable energy solutions, particularly in densely built urban environments where conventional solar installations are limited. **Building Integrated Photovoltaic (BIPV)** systems offer a promising solution by enabling energy generation directly from building facades and rooftops. In this context, accurate estimation of solar energy potential at the building level is essential for effective planning and implementation of BIPV systems.

This internship work focuses on “**3D Urban Building Energy Assessment for BIPV Applications using Geospatial Techniques**”, carried out at **Indian Space Research Organisation (ISRO) – Space Applications Centre (SAC), Ahmedabad**. The objective of the project is to develop a geospatial workflow for assessing solar energy potential of urban buildings using three-dimensional (3D) building data, while accounting for shadow effects caused by surrounding structures.

The methodology involves processing geospatial building datasets to generate 3D urban models with height information. Shadow analysis and solar radiation estimation are performed using computational techniques, with parallelized processing to improve efficiency. The processed outputs are visualized through interactive 3D models and geospatial visualizations, enabling detailed analysis of building-wise energy potential. The results are exported in standard formats suitable for visualization and further analysis.

The developed framework demonstrates the effectiveness of integrating geospatial techniques, 3D modeling, and solar energy analysis for urban energy assessment. This work highlights the potential application of the proposed approach in urban planning, renewable energy deployment, and smart city initiatives. The outcomes of this internship contribute towards efficient assessment of BIPV feasibility and provide a foundation for future research in the integration of **geospatial analytics, artificial intelligence, and large-scale urban energy modeling**.

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- **This index provides a structured overview of the content of the ISRO internship report. Each section is meticulously organized to guide the reader through the report's comprehensive analysis and findings.**

CHAPTER 1 – INTRODUCTION

Rapid urbanization and population growth have significantly increased the demand for energy in urban areas, leading to greater dependence on conventional energy sources and increased environmental impact. In response to global concerns regarding climate change and sustainable development, there is a growing emphasis on the adoption of renewable energy resources, particularly solar energy. Urban environments, however, pose unique challenges for solar energy deployment due to limited open spaces, complex building geometries, and shadowing effects caused by surrounding structures.

Building Integrated Photovoltaic (BIPV) systems provide an effective solution by enabling solar energy generation directly from building envelopes such as rooftops and facades. The feasibility and efficiency of BIPV installations depend heavily on accurate assessment of solar energy potential at the building level. Traditional two-dimensional (2D) assessment methods are often insufficient in dense urban settings, as they fail to account for building height variations, orientation, and shadow interactions. Consequently, there is a strong need for **three-dimensional (3D) urban energy assessment frameworks** that incorporate geospatial data and advanced computational techniques.

Geospatial technologies play a crucial role in urban energy modeling by enabling spatial analysis, data integration, and visualization of complex urban datasets. With the availability of 3D city models and building-level geospatial data, it has become possible to perform detailed solar radiation and shadow analysis for individual buildings. However, large-scale implementation of such analyses requires efficient data processing pipelines, robust computational methods, and scalable visualization frameworks.

This internship project, titled “**3D Urban Building Energy Assessment for BIPV Applications using Geospatial Techniques**”, was carried out at **Indian Space Research Organisation (ISRO) – Space Applications Centre (SAC), Ahmedabad**. The primary objective of this work is to develop a geospatial and computational workflow for assessing the solar energy potential of urban buildings using 3D building data, while considering shadow effects and urban morphology.

The Level of Detail 1 (LOD1_Building_with_height_info_8buildings) building dataset used in this study was prepared using the deep learning methodology proposed by **Rastogi et al. (2022)** for the automatic extraction of building footprints from very high-resolution satellite imagery. The data preparation involved using 4-band pan-sharpened Cartosat-2 series imagery with a ground resolution of less than 1 meter to ensure high spatial accuracy. To create a reliable

training set, existing OpenStreetMap (OSM) vector data was manually georeferenced and corrected to match the building shapes in the satellite images before being converted into binary raster masks. A specialized Convolutional Neural Network (CNN) architecture, known as UNet-AP, was then employed to perform pixel-wise labeling and delineate precise building boundaries across various urban settlement types. The final footprints were converted into GIS-compatible polygons, with erroneous small structures filtered out to ensure a high-quality dataset suitable for urban-scale analysis.

The computational methodology integrates multiple geospatial and scientific computing tools. Building geometries are processed using **Python-based GIS libraries**, including **GeoPandas**, **Shapely**, **pyshp**, and **pyproj**, for spatial data handling and coordinate transformations. Numerical computations and data management are performed using **NumPy** and **Pandas**, enabling efficient handling of large datasets. Solar position, radiation, and energy estimation are carried out using **pvlib**, while shadow analysis and ray-based computations utilize **pyembree** for improved performance.

To support three-dimensional visualization and interactive analysis, building models are generated and exported using **Trimesh** and related 3D processing tools. The output models are visualized through web-based platforms using **CesiumJS**, along with **HTML**, **CSS**, and **JavaScript (JSON-based data handling)**. Additionally, **QGIS** is used for spatial data inspection, preprocessing, and validation. This integrated toolchain enables both quantitative analysis and intuitive visualization of building-wise solar energy potential.

The computation focuses on both **shadow-aware solar radiation analysis** and **solar energy potential estimation**, providing insights into the suitability of individual building surfaces for BIPV deployment. Parallelized processing techniques are employed to enhance computational efficiency, making the framework suitable for large-scale urban applications.

The significance of this work lies in its potential applications in **urban planning, renewable energy deployment, smart city initiatives, and policy-level decision-making**. By providing building-level energy assessment, the developed framework can assist stakeholders in identifying optimal locations for BIPV installations and improving the utilization of available urban surfaces for solar energy generation.

Furthermore, this project establishes a foundation for future research in the integration of **artificial intelligence, machine learning, big data, and geospatial modeling** for large-scale urban energy assessment. The proposed framework can be extended to incorporate machine learning models for predictive analysis, and city-wide energy optimization, thereby contributing to advanced research and development in the field of geospatial energy analytics.

CHAPTER 2 - ORGANIZATION PROFILE

The **Indian Space Research Organisation (ISRO)** is the national space agency of India, functioning under the **Department of Space (DoS), Government of India**. Established in 1969, ISRO is responsible for the development and application of space technology for national development, scientific research, and exploration of outer space. Over the decades, ISRO has made significant contributions in the fields of satellite technology, launch vehicle development, space science, and space-based applications for societal benefit.

ISRO's mission focuses on harnessing space technology for communication, navigation, earth observation, disaster management, meteorology, environmental monitoring, and sustainable development. Through its extensive satellite programs and ground infrastructure, ISRO supports a wide range of civilian and strategic applications, contributing to national growth and technological self-reliance.

The **Space Applications Centre (SAC)**, located in Ahmedabad, Gujarat, is one of ISRO's major research and development centres. SAC is primarily responsible for the development of **spaceborne payloads, satellite-based applications, and advanced technologies** related to earth observation, communication, navigation, and meteorology. The centre plays a crucial role in translating space technology into practical applications for societal, environmental, and economic development.

SAC is actively involved in the design and development of payloads for **remote sensing satellites, meteorological satellites, and navigation missions**. Its research activities include microwave and optical remote sensing, geospatial data processing, image analysis, atmospheric studies, and development of application-oriented tools for agriculture, urban planning, water resources, climate studies, and disaster management.

In addition to payload development, SAC contributes significantly to the advancement of **geospatial technologies and data analytics**. The centre focuses on developing methodologies for processing large volumes of satellite data, generating geospatial products, and delivering decision-support systems to various user agencies. These activities support national initiatives such as **urban development, renewable energy planning, smart cities, and environmental sustainability**.

SAC also emphasizes research and development in emerging domains, including **3D geospatial modeling, data-driven analytics, and advanced visualization techniques**. By integrating satellite data, geographic information systems (GIS), and computational methods, the centre enables detailed analysis of complex urban and environmental systems.

The internship and research training programs at SAC provide students and researchers with exposure to a professional research environment, allowing them to work on real-world problems under the guidance of experienced scientists and engineers. These programs foster technical skill development, research aptitude, and an understanding of space-based applications in addressing national challenges.

Overall, the Space Applications Centre (SAC) plays a vital role in ISRO's mission by bridging space technology and its practical applications, thereby contributing to scientific advancement, technological innovation, and sustainable national development.

CHAPTER 3 - PROBLEM STATEMENT & OBJECTIVES

Urban environments present significant challenges for the effective utilization of solar energy due to dense building configurations, complex geometries, and mutual shadowing effects. Although **Building Integrated Photovoltaic (BIPV)** systems offer an efficient approach for harvesting solar energy from building rooftops and facades, their successful implementation requires accurate, building-level assessment of solar energy potential using three-dimensional (3D) data.

In the present project, an initial geospatial and 3D visualization pipeline had been developed to generate urban building model and visualize them using **CesiumJS**. While the pipeline successfully rendered building rooftops and facades, it exhibited **critical functional and computational limitations** that restricted its practical usability and scalability.

One of the major issues observed was the **absence of interactive data binding between the 3D building geometry and the underlying solar energy attributes**. Although building surfaces were visually displayed in CesiumJS, user interaction such as clicking or inspecting individual building components failed to return any associated solar or BIPV-related information. The inspection process resulted in errors, indicating broken or incomplete linkage between the generated 3D models and their corresponding analytical metadata. This significantly limited the usefulness of the visualization framework for decision-making and analysis.

A second major challenge was the **high computational time of the processing python code pipeline**. For a pilot dataset consisting of only **eight buildings**, the execution time of the pipeline was approximately **~ 32 minutes**. Such performance raised serious concerns regarding scalability, particularly when considering city-scale deployment for **thousands of buildings across Ahmedabad**. The lack of automation, redundant computations, and sequential processing significantly increased the overall time complexity of the system.

Furthermore, the original pipeline lacked efficient resource utilization and dynamic adaptation to available system hardware. The absence of parallel processing and caching mechanisms resulted in repeated calculations, increased computational overhead, and underutilization of available CPU resources.

Therefore, the core problem addressed in this internship was the need to **redesign, optimize, and automate a 3D urban solar energy assessment pipeline**. Addressing these challenges is essential for enabling **scalable, interactive, and data-driven urban BIPV assessment using**

geospatial techniques, supporting sustainable urban planning and renewable energy deployment.

The primary objective of this internship is to **design, optimize, and implement a scalable geospatial framework for 3D urban building energy assessment focused on Building Integrated Photovoltaic (BIPV) applications.**

The specific objectives are:

- 1. Automated Mesh-to-Metadata Generation** for high-resolution BIPV assessment using LOD1 building data.
- 2. Shadow-Aware BIPV Energy Modeling** using ray tracing techniques to incorporate realistic shading effects.
- 3. Scalable Urban Solar Data Pipeline** with computational optimization to reduce execution time and improve efficiency.
- 4. 3D JSON Tileset Visualization** using CesiumJS integrated with BIPV energy information for interactive urban-scale analysis.

CHAPTER 4 - LITERATURE REVIEW

Urbanization and the rapid growth of built-up areas have significantly increased the demand for sustainable and decentralized energy solutions. In this context, **Building Integrated Photovoltaic (BIPV) systems** have emerged as a promising approach for utilizing urban building surfaces for renewable energy generation (**Jelle et al., 2012**). Accurate assessment of BIPV potential requires detailed **spatial, geometric, and solar radiation analysis**, which has been addressed by several researchers using **geospatial and computational techniques**.

Early studies on urban solar potential primarily relied on **two-dimensional (2D) Geographic Information System (GIS)** data and digital elevation models (DEMs) to estimate rooftop solar energy potential. While these approaches provided initial insights, they were limited in their ability to account for **building height variations, façade orientation, and inter-building shadow effects**, which are critical in dense urban environments.

With advancements in **3D city modeling standards**, such as **Level of Detail (LOD)** representations, researchers began utilizing LOD1 and LOD2 building models to improve the accuracy of solar energy estimation. LOD1 models, which include building footprints with height information, have been widely adopted due to their availability and computational efficiency for large-scale urban studies. However, the accurate integration of LOD1 models with solar radiation algorithms remains an active research area.

Several studies have employed **solar position algorithms and radiative transfer models** to estimate incident solar energy on building surfaces. Libraries and models based on astronomical calculations have enabled precise computation of solar angles and irradiance values. Nevertheless, these methods often face challenges related to **computational complexity**, especially when applied to large urban datasets with fine temporal resolution.

Shadow analysis plays a critical role in urban solar energy assessment. Researchers have explored **ray-tracing techniques**, voxel-based methods, and geometry-based intersection models to quantify shading effects caused by neighboring buildings. While these methods improve accuracy, they significantly increase computational cost, making optimization and parallel processing essential for scalable implementations.

Recent advancements have focused on integrating **Python-based geospatial libraries**, such as GeoPandas, Shapely, and numerical computing tools, to automate urban solar potential workflows. Additionally, the use of **web-based 3D visualization platforms**, including WebGL-based engines, has enabled interactive exploration of urban energy datasets. However,

effective linkage between computational results and real-time 3D visualization remains a challenge.

Despite these advancements, existing literature highlights a gap in **scalable, optimized, and visualization-ready frameworks** that combine shadow-aware energy estimation with interactive 3D urban models. This internship work aims to address these limitations by developing an optimized geospatial pipeline that integrates solar computation, parallel processing, and interactive 3D visualization for BIPV-focused urban energy assessment.

CHAPTER 5 – METHODOLOGY AND SYSTEM ARCHITECTURE

The methodology adopted in this project focuses on the development of a scalable, automated, and computationally efficient framework for **3D urban building energy assessment for Building Integrated Photovoltaic (BIPV) applications** using geospatial techniques. The approach integrates **GIS-based spatial data processing, solar radiation modeling, shadow-aware energy computation, and web-based 3D visualization** into a unified pipeline.

The workflow begins with the ingestion of **LOD1 building footprint data with height attributes**, followed by the generation of **3D building geometries**. Each building is decomposed into roof and façade elements to enable surface-wise solar and energy analysis. Advanced computational techniques such as **parallel processing, solar position caching, and scene-level ray tracing** are employed to improve execution efficiency and scalability. The final outputs are rendered as **interactive 3D models in a CesiumJS environment**, enabling façade-level inspection of BIPV potential.

A high-level representation of the complete workflow is illustrated in **Figure 5.1**.

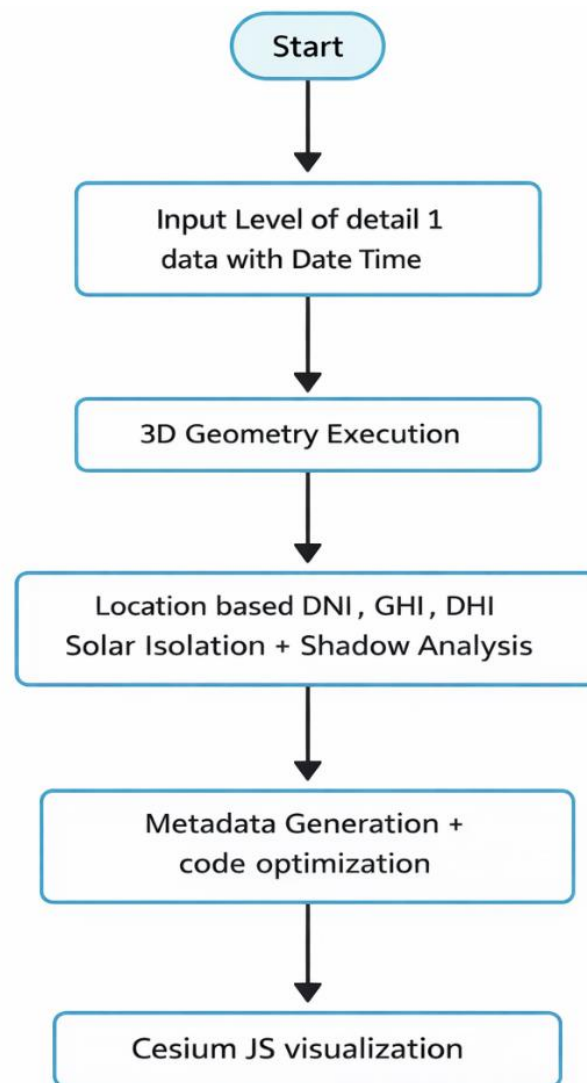
5.1 Overall Workflow

The developed system follows a modular and automated workflow designed to ensure scalability, computational efficiency, and ease of visualization. The major stages of the workflow are as follows:

1. Generation of three-dimensional building geometry
2. Shadow-aware solar radiation and energy computation
3. Optimization through Mesh Combining and caching
4. Generation of visualization-ready outputs
5. Interactive 3D visualization using web-based platforms

Each stage of the workflow is designed to function independently, allowing flexibility for future enhancements and large-scale deployment.

Figure – 5.1



5.2 Input Data and Study Area

The primary data source used in this study consists of building footprints extracted from **Cartosat-2 series 4-band multispectral satellite imagery** with less than 1-meter resolution. The footprints were automatically delineated using a novel deep learning architecture, **UNet-AP**, which utilizes **Atrous Spatial Pyramid Pooling (ASPP)** to accurately capture building boundaries across diverse urban settlements. The resulting dataset was converted into a **vector shapefile format** with planar footprints and vertical height attributes suitable for LOD1 3D modeling. The dataset includes planar building footprints

along with vertical height attributes, which are sufficient for generating simplified 3D building models suitable for large-scale urban analysis.

The **study area** currently consists of **eight representative buildings located in Ahmedabad, Gujarat**, selected for pipeline development and validation. Although the present implementation focuses on a limited subset, the system architecture and computational design are explicitly developed to support **city-scale processing for the entire Ahmedabad urban region** in future deployments.

All spatial data are transformed and processed in **EPSG: 32643 (UTM Zone 43N)** coordinate reference system to ensure spatial accuracy and geometric consistency. This projection is suitable for the Ahmedabad region and preserves metric distances and areas, which is essential for precise solar radiation computation and façade-level BIPV energy estimation. Using a projected coordinate system also enables accurate surface area calculations and reliable shadow and orientation analysis for urban-scale energy modelling.

5.3 3D Building Geometry Generation

Following data ingestion, each building footprint is extruded vertically using height attributes to construct a **3D building model**. The geometry generation process includes:

- Identification of roof surfaces and vertical façade surfaces
- Computation of surface normals for orientation analysis
- Subdivision of large façades into smaller analytical units (sub-facades)
- Calculation of centroid coordinates and surface areas

This decomposition enables fine-grained solar and energy estimation at the façade level, which is critical for realistic BIPV feasibility assessment.

5.4 Solar Radiation and Energy Modeling

Solar radiation modeling is performed using the **pvlib Python library**, which provides physically based solar position and irradiance calculations. For a selected analysis date, the following steps are executed:

- Solar position computation (azimuth, zenith, elevation)
- Clear-sky irradiance estimation using the simplified Solis model

- Plane-of-array (POA) irradiance calculation for roofs and façades
- Orientation-aware irradiance adjustment based on surface tilt and azimuth

Solar radiation and shadow computations are performed at hourly intervals to maintain sufficient temporal resolution for accurate daily energy estimation. Solar positions are further optimized using a **global caching mechanism**, ensuring reuse of solar calculations for nearby façades with similar geographic locations.

5.5 Shadow Analysis using Ray Tracing

Accurate shadow detection is essential for urban BIPV assessment due to obstruction effects from neighboring structures. In this project, **ray tracing-based shadow analysis** is implemented using the **Trimesh library**.

A global 3D scene comprising all building surfaces is constructed, and **batch ray casting** is performed for each façade and time step. Rays are projected from the sun position toward façade centroids to determine whether direct solar radiation is obstructed. Shadow effects are incorporated by suppressing direct irradiance components for shadowed time intervals.

This shadow-aware approach significantly improves the realism of solar and energy estimates in dense urban environments.

5.6 Performance Optimization and Parallel Processing

One of the major challenges addressed in this project is **computational scalability**. Initial pipeline execution required approximately **~32 minutes for processing eight buildings**, indicating infeasibility for city-scale analysis.

To overcome this limitation, several optimization strategies were introduced:

- **Building-level parallel processing** using Python multiprocessing
- Automatic worker allocation based on system CPU availability
- Scene-level acceleration structure (BVH) precomputation
- Removal of redundant computations and unused code blocks
- Efficient batching of ray tracing operations

As a result, pipeline execution time was reduced progressively from **32 minutes to approximately 3 minutes**, achieving nearly **11× performance improvement without GPU acceleration**. The framework is designed to support future GPU-based acceleration for further speedups.

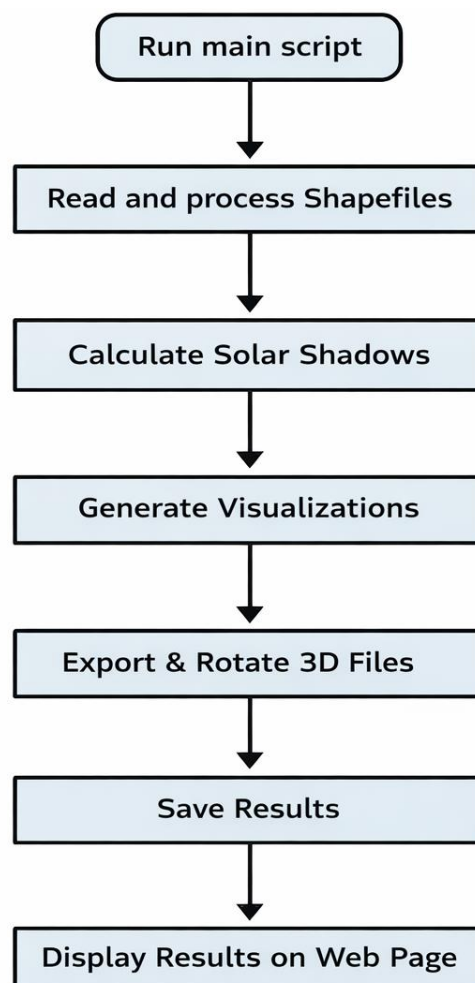
5.7 Output Generation and Data Structuring

The processed results are exported in two primary formats:

- **GLB (GL Transmission Format)** for 3D visualization of buildings with color-coded BIPV energy density
- **Anchor JSON files** containing façade-wise metadata such as solar potential, BIPV potential, energy density, sunshine hours, and shadow fractions

Each façade is linked to its corresponding metadata, enabling dynamic retrieval of energy information during user interaction in the visualization environment.

Figure – 5.8



5.9 System Architecture Overview

The complete system architecture follows a **layered design approach**, consisting of:

- **Input Data Layer:** LOD1 shapefiles with height attributes
- **Processing Layer:** Python-based geospatial, solar, and energy computation modules
- **Optimization Layer:** Parallel execution engine and caching mechanisms
- **Output Layer:** GLB models and structured JSON metadata
- **Visualization Layer:** Web-based CesiumJS interactive 3D viewer

This modular architecture ensures flexibility, scalability, and ease of future enhancements. A schematic representation of the system architecture is shown in **Figure 5.8**

Summary

This chapter presented a comprehensive description of the methodology and system architecture developed for 3D urban BIPV energy assessment. By integrating geospatial data processing, solar radiation modeling, shadow-aware analysis, and performance optimization techniques, the proposed framework demonstrates both **technical robustness and practical scalability**. The modular design and optimized execution pipeline make the system suitable for extending from building-level analysis to **city-scale urban solar energy planning**.

CHAPTER 6 - RESULTS AND DISCUSSION

6.1 Overview of Results

This chapter presents the results obtained from the implementation of the proposed **3D Urban Building Energy Assessment framework for BIPV applications**. The outcomes include processed three-dimensional building models, surface-wise solar and BIPV energy metrics, shadow impact assessment, and performance-related results achieved through computational optimization.

The results demonstrate the effectiveness of the developed pipeline in producing accurate, scalable, and visualization-ready outputs suitable for urban solar energy analysis.

6.2 Generated Output Files

The execution of the pipeline produces two primary categories of outputs, organized systematically to support efficient visualization and analysis:

6.2.1 3D Geometry Output (GLB Files)

The three-dimensional building models are exported in **GLB format**, which is compatible with modern web-based visualization platforms such as CesiumJS. These files contain:

- Extruded LOD1 building geometry
- Correct spatial orientation and alignment
- Surface-wise color encoding based on BIPV energy density

The GLB files allow real-time rendering and interactive exploration of urban buildings in a geospatial environment.

6.2.2 Analytical Metadata Output (Anchor JSON Files)

The anchor files are generated in **JSON format** and store detailed analytical information for each building surface, including:

- Sunshine hours
- Solar energy potential (kWh)
- BIPV potential (kWh)
- BIPV energy density (kWh/m²)
- Shadow fraction
- Time-wise shadow information

This separation of geometry and metadata enables lightweight data handling and dynamic querying during visualization.

6.3 Solar and BIPV Energy Assessment Results

Solar energy assessment was performed for both rooftop and façade surfaces of the selected buildings. The results indicate clear variation in solar exposure based on surface orientation, height, and surrounding obstructions.

Key observations include:

- Rooftop surfaces generally exhibit higher solar potential due to minimal shadow obstruction.
- Façade surfaces show orientation-dependent energy variation, with south- and west-facing façades yielding higher BIPV density.
- Shadow-aware computation significantly influences energy estimation, preventing overestimation commonly observed in shadow-ignorant models.

The calculated **BIPV energy density (kWh/m²)** enables surface-level comparison, which is essential for practical BIPV planning.

6.4 Shadow Impact Analysis Results

Shadow analysis plays a critical role in realistic urban solar assessment. The implemented ray-tracing-based approach identifies shadowed and unshaded periods for each surface.

The results reveal:

- Temporal variation in shadow coverage throughout the day

- Increased shadow fraction for lower façades and closely spaced buildings
- Reduction in effective solar irradiance due to partial or complete shading

The shadow fraction values computed for each surface directly influence the final BIPV energy estimation, ensuring physical realism in dense urban environments.

6.5 Computational Performance Results

A significant outcome of this internship is the **substantial improvement in computational efficiency** achieved through code optimization and parallel processing.

6.5.1 Execution Time Reduction

Initial execution time for processing three buildings using the earlier implementation was approximately **32 minutes**. Through systematic optimization, the execution time was reduced in multiple stages:

Stage	Approximate Time
Initial implementation	~32 minutes
Code automation & cleanup	~16 minutes
Algorithm optimization	~8 minutes
Combined Mesh implementation	~3 minutes

These improvements were achieved **without GPU acceleration**, demonstrating the effectiveness of CPU-based optimization strategies.

6.5.2 Optimization Impact

The reduction in execution time was achieved through:

- Building-level parallel processing
- Solar position caching
- Reduced temporal sampling

- Efficient batch ray tracing
- Dynamic worker allocation based on system resources

This performance enhancement confirms the feasibility of scaling the framework to handle larger urban datasets.

6.6 Validation of Results

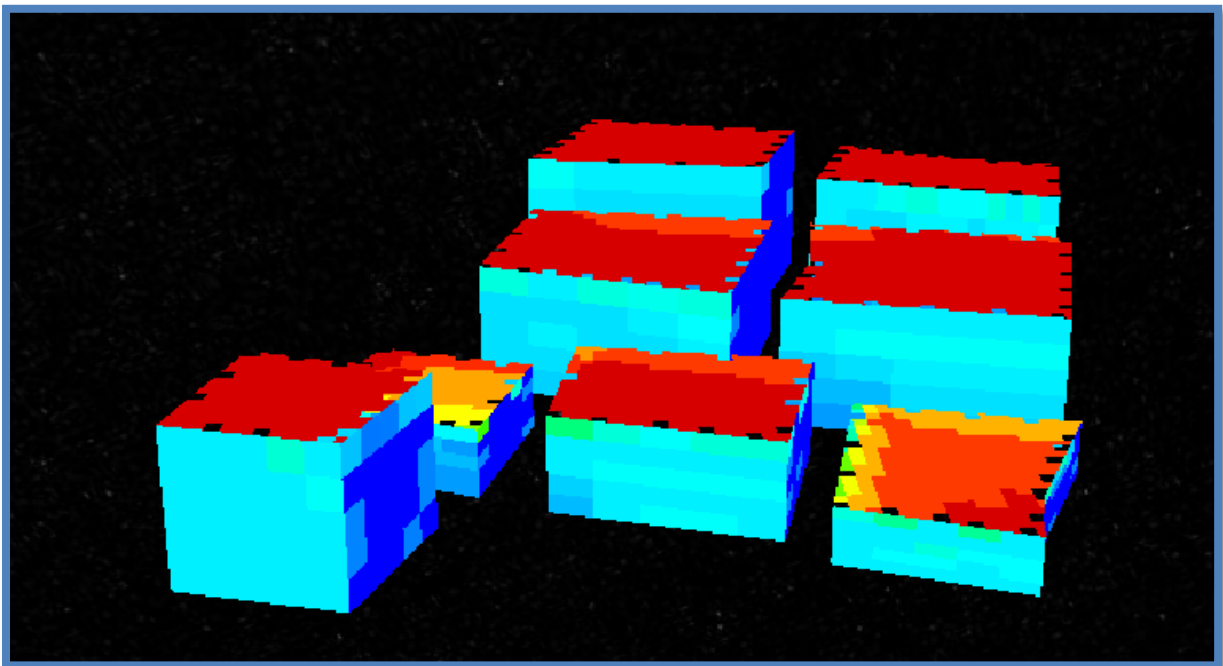
The generated outputs were validated through:

- Visual inspection of 3D models in the CesiumJS environment
- Cross-verification of energy values across different surfaces
- Consistency checks between geometry and metadata

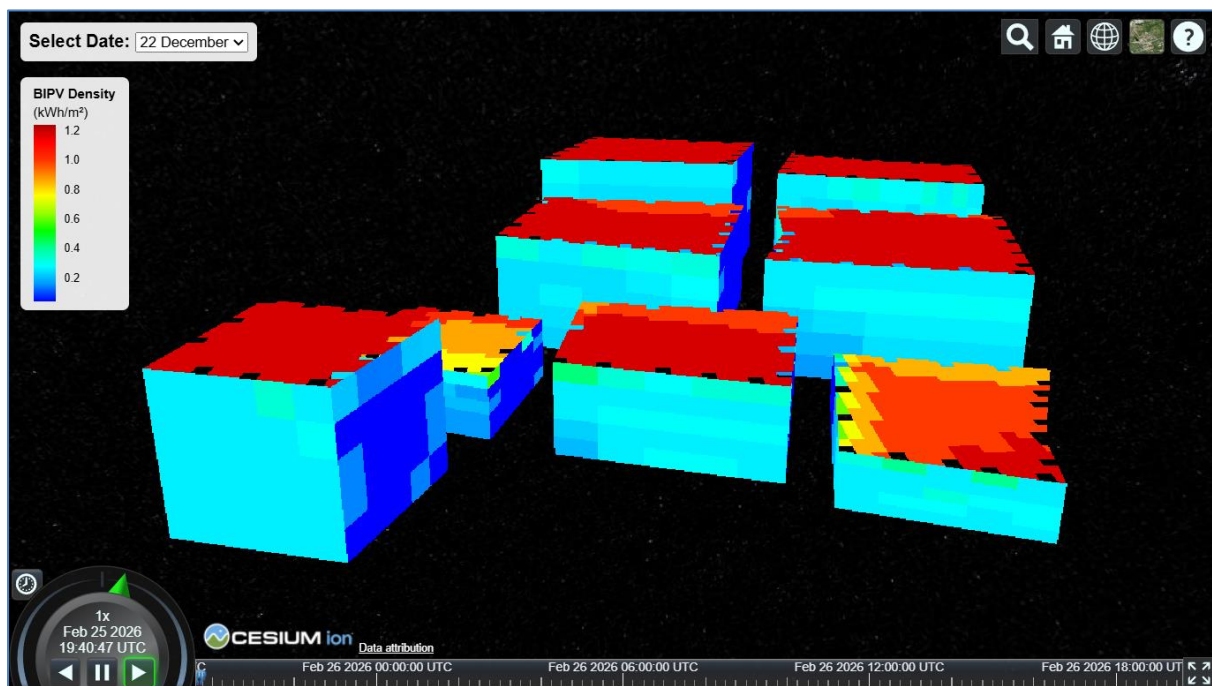
The successful interaction between GLB models and anchor metadata confirms the correctness of the output structure and data linkage.

This chapter presented the results obtained from the developed 3D urban solar energy assessment framework. The outputs demonstrate accurate geometry generation, realistic shadow-aware solar and BIPV energy estimation, and significant improvements in computational performance. The generated results form a strong foundation for interactive visualization and practical BIPV decision-making, which are discussed in the subsequent chapter.

6.7. EIGHT BUILDINGS IN CESIUM – WITHOUT CLICK



6.8. User Interface demonstrating subfacade level BIPV for 8 buildings



6.9. User Interface demonstrating on click metadata information

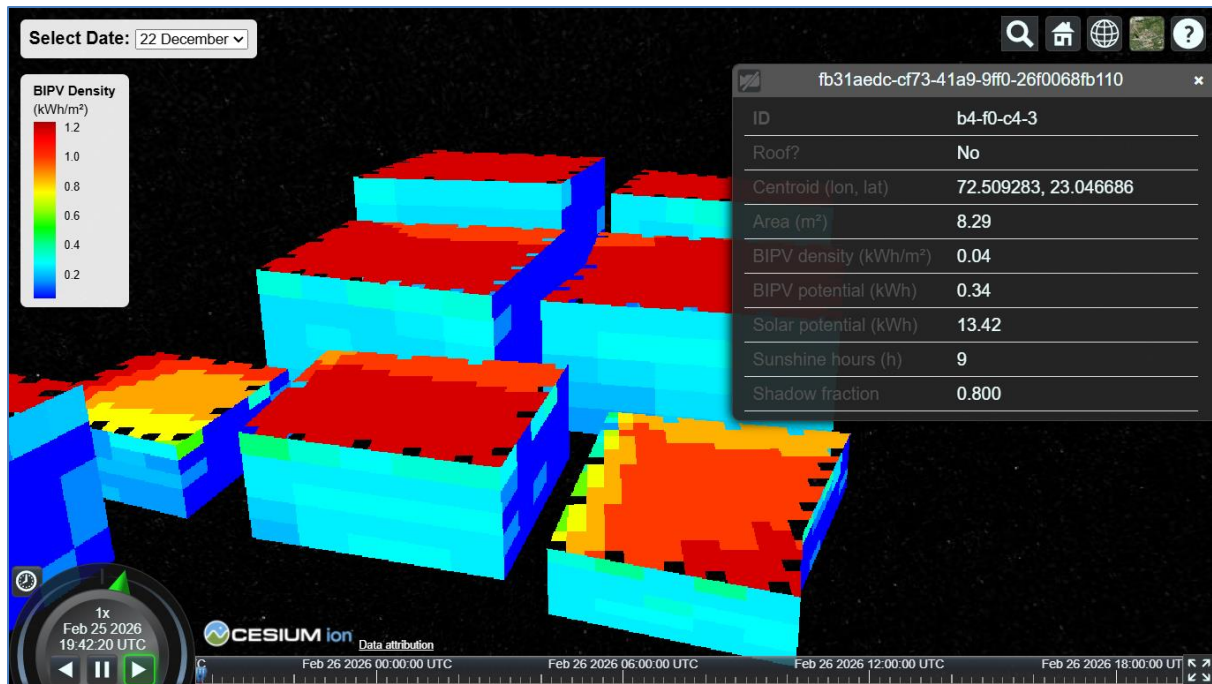


Figure 6.9 illustrates the interactive three-dimensional visualization of urban buildings generated as part of the solar energy assessment framework. The visualization is rendered using **CesiumJS**, where each building is represented as a Level of Detail 1 (LOD1) 3D model derived from geospatial building footprint data and height information.

The rooftop and façade surfaces of the buildings are color-coded based on **Building Integrated Photovoltaic (BIPV) energy density (kWh/m^2)**. A continuous color scale is used, ranging from low potential (blue) to high potential (red), enabling intuitive interpretation of solar suitability across different building surfaces.

The interface allows **temporal selection**, as shown by the date selector, enabling analysis of solar energy potential for specific days. Solar position and shadow effects are dynamically reflected in the visualization, ensuring that energy values account for **shadow fraction**, **sunshine hours**, and **solar irradiance** at the selected time.

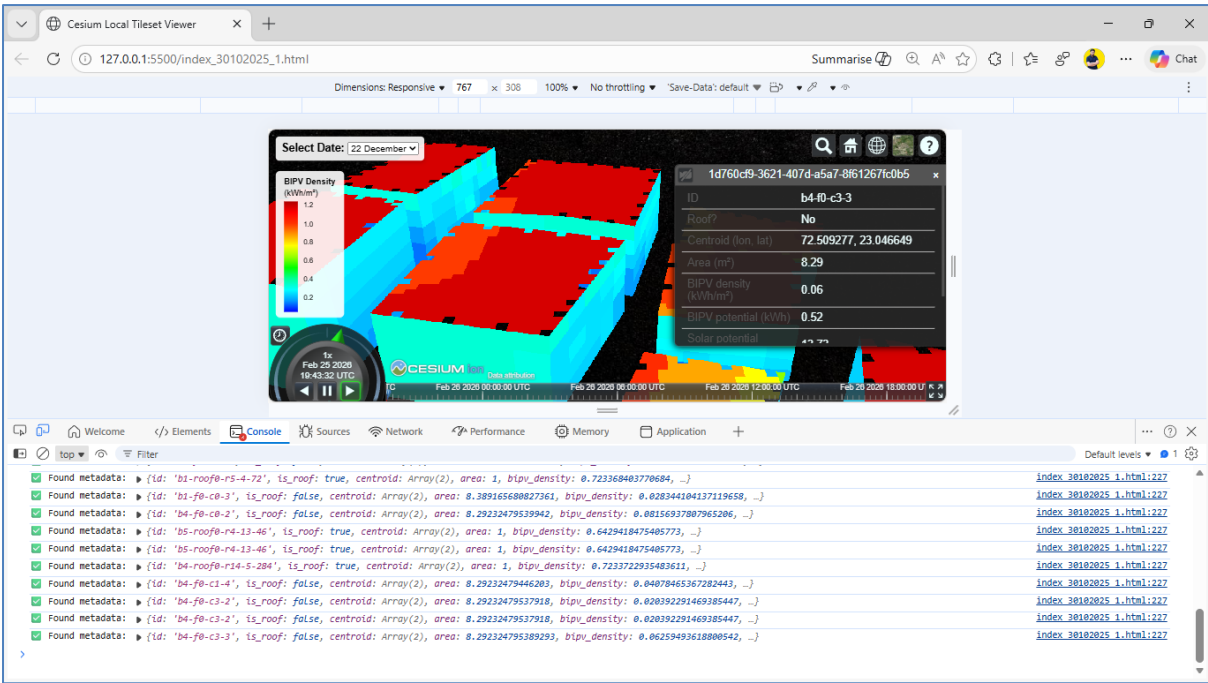
On selecting an individual surface element, a detailed information panel is displayed, providing quantitative attributes such as:

Surface identification and roof classification

- Geographic centroid (latitude and longitude)
- Surface area (m^2)
- BIPV energy density (kWh/m^2)
- Estimated BIPV energy potential (kWh)
- Total solar potential (kWh)
- Sunshine duration (hours)
- Shadow fraction

This interactive visualization enables building-level and surface-level analysis of solar energy potential, supporting informed decision-making for **urban BIPV deployment**. The integration of geospatial accuracy, shadow-aware computation, and web-based 3D visualization demonstrates the effectiveness of the developed framework for scalable urban energy assessment.

6.10. Metadata generation



6.11. JSON file containing anchor

```
3buildings_optimized_anchor.json X
output > anchor > {} 3buildings_optimized_anchor.json > {} crs
1 {
2   "longitude": 72.50951981447494,
3   "latitude": 23.04675642425395,
4   "height": 5.336622939614899,
5   "utm_anchor": {
6     "x": 244883.6657938886,
7     "y": 2558867.9197728713,
8     "z": 5.336622939614899
9   },
10  "crs": {
11    "source": "EPSG:32643",
12    "target": "EPSG:4326"
13  },
14  "items": {
15    "b0-f0-c0-0": {
16      "id": "b0-f0-c0-0",
17      "is_roof": false,
18      "centroid": [
19        72.50971924793575,
20        23.046834633770388
21      ],
22      "area": 0.9716377799533403,
23      "bipv_density": 0.0,
24      "bipv_potential": 0.0,
25      "solar_potential": 0.7812096551730187,
26      "sunshine_hours": 0,
27      "shadow_fraction": 1.0
28    },
29    "b0-f0-c0-1": {
30      "id": "b0-f0-c0-1",
31      "is_roof": false,
32      "centroid": [
33        72.50971924793575,
34        23.046834633770388
35      ],
36      "area": 0.9716377799533403,
37      "bipv_density": 0.0,
38      "bipv_potential": 0.0,
39      "solar_potential": 0.7812096551730187,
40      "sunshine_hours": 0,
41      "shadow_fraction": 1.0
42    }
43  }
44 }
```

The Figure 6.11 shows anchor metadata file stores surface-wise solar and geometric information for each building in a structured JSON format. It includes geographic coordinates, height, coordinate reference system (CRS) details, and centroid locations to ensure spatial accuracy. For every roof and façade surface, attributes such as area, BIPV energy density, solar potential, sunshine hours, and shadow fraction are recorded.

CHAPTER 7 - APPLICATIONS AND FUTURE SCOPE

7.1 Applications of the Developed Framework

The developed **3D Urban Building Energy Assessment framework for BIPV applications** has wide applicability in urban planning, renewable energy assessment, and smart city initiatives. By providing surface-level solar and BIPV energy insights, the framework supports informed decision-making across multiple domains.

7.1.1 Urban Energy Planning

The system can be used by urban planners and local authorities to identify high-potential rooftops and façades for solar energy deployment. The availability of façade-level analysis enables more effective utilization of vertical surfaces in dense urban environments where rooftop space is limited.

7.1.2 Building Integrated Photovoltaic (BIPV) Deployment

The framework assists architects and engineers in evaluating the suitability of building surfaces for BIPV installation during both new construction and retrofit planning. Surface-wise energy density values enable optimized placement of photovoltaic panels, improving overall energy yield.

7.1.3 Smart City and Sustainability Initiatives

The interactive 3D visualization supports smart city applications by integrating renewable energy assessment with geospatial decision-support systems. The framework can aid policymakers in developing sustainable urban energy strategies and monitoring renewable energy potential at the city scale.

7.2 Future Scope of Work

Although the current implementation successfully demonstrates the feasibility of 3D urban BIPV assessment, several enhancements can be explored in future work to improve accuracy, scalability, and functionality.

7.2.1 City-Scale and Regional Expansion

The framework can be extended to process the complete building inventory of Ahmedabad and other cities, enabling comprehensive urban-scale solar energy assessment.

7.2.2 GPU Acceleration and High-Performance Computing

Future implementations may integrate GPU-based ray tracing and numerical computation to significantly reduce execution time for large datasets. This enhancement would support real-time or near-real-time city-scale analysis.

7.2.3 Integration with Artificial Intelligence and Big Data

Machine learning techniques can be incorporated to:

- Predict solar potential using historical data
- Automatically classify building surfaces based on BIPV suitability
- Optimize panel placement and configuration

Big data platforms can further enhance scalability and data management.

7.2.4 Higher Level of Detail (LOD) Modeling

Incorporating LOD2 or LOD3 building datasets with detailed roof structures and façade elements can improve accuracy in shadow modeling and energy estimation.

7.3 Summary

This chapter discussed the practical applications of the developed framework and outlined potential directions for future research and development. The proposed enhancements can further strengthen the framework's role in urban renewable energy planning and advanced geospatial research.

CHAPTER 8 - CONCLUSION

This internship works successfully presented a **3D Urban Building Energy Assessment framework for Building Integrated Photovoltaic (BIPV) applications using geospatial techniques**. The study demonstrated how three-dimensional urban building data, when integrated with shadow-aware solar energy computation and interactive visualization, can provide meaningful insights into urban renewable energy potential.

The developed methodology effectively combined **LOD1 building datasets**, Python-based geospatial processing, and solar energy modeling to estimate rooftop and façade-level energy potential. By incorporating shadow analysis and temporal solar position calculations, the framework addressed key challenges associated with dense urban environments, where mutual shading significantly impacts solar performance.

The system architecture adopted a **modular and scalable design**, enabling independent processing of data acquisition, geometry generation, solar computation, optimization, and visualization. The separation of **geometry (GLB)** and **metadata (anchor JSON)** ensured efficient data handling and seamless integration with web-based 3D visualization platforms such as **CesiumJS**.

The interactive visualization environment enhanced the interpretability of results by allowing users to explore building-wise energy values, shadow effects, and BIPV suitability in a realistic geospatial context. This approach highlights the importance of visual analytics in supporting urban energy planning and decision-making.

Although the current implementation focused on a limited set of buildings in Ahmedabad as a pilot study, the framework is inherently extensible to city-scale applications. With future enhancements such as higher **Levels of Detail (LOD)**, **high-performance computing**, and **artificial intelligence integration**, the system has strong potential for large-scale deployment in smart city and sustainability initiatives.

Overall, this internship provided valuable exposure to **geospatial analytics, urban energy modeling, and applied research workflows** at the **Indian Space Research Organisation (ISRO)**. The outcomes of this work demonstrate the relevance of geospatial techniques in addressing real-world energy challenges and contribute towards the broader objective of sustainable and intelligent urban development.

CHAPTER 9 - REFERENCES

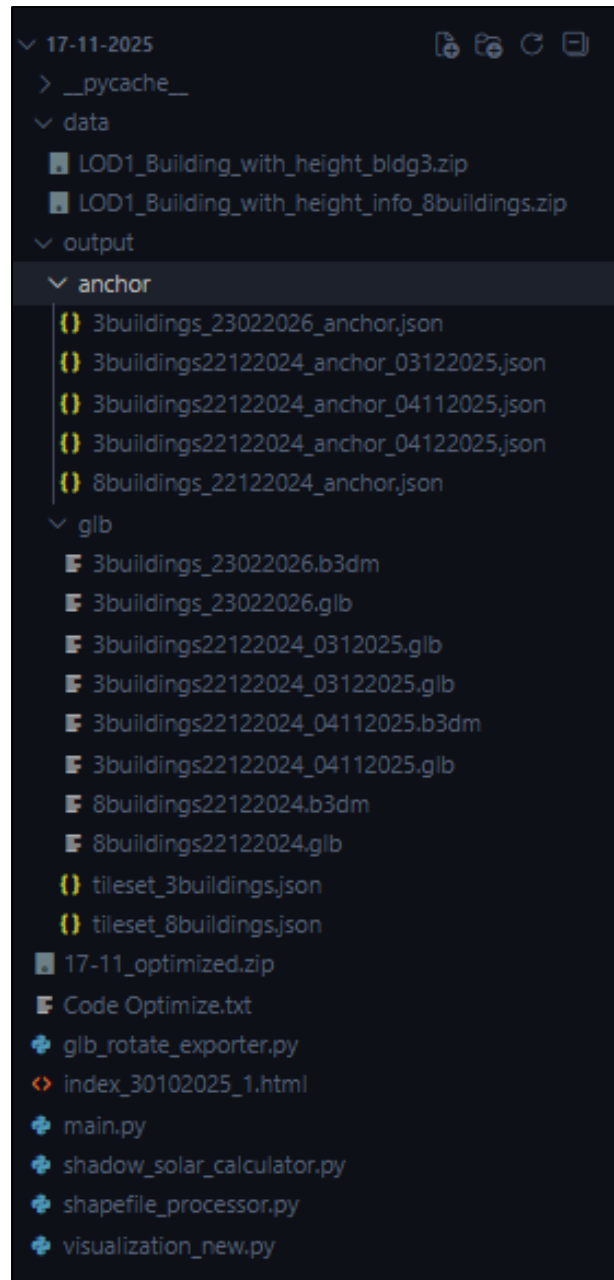
1. LOD – 1 Reference - Kriti Rastogi, Pankaj Bodani & Shashikant A. Sharma (2022) Automatic building footprint extraction from very high-resolution imagery using deep learning techniques, Geocarto International, 37:5, 1501-1513, DOI: 10.1080/10106049.2020.1778100
2. Jelle, B. P., Breivik, C., & Røkenes, H. D. (2012). Building integrated photovoltaics: A review. *Solar Energy Materials and Solar Cells*, 100, 69–96. <https://doi.org/10.1016/j.solmat.2011.12.016>
3. Python – [Our Documentation | Python.org](#)

4. GeoPandas - [GeoPandas Tutorial - GeeksforGeeks](#)
5. Shapely – [shapely · PyPI](#)
6. NumPy – [NumPy - Learn](#)
7. Pandas – [Pandas Tutorial](#)
8. pvlib – [pvlib python 0.15.0 documentation](#)
9. Trimesh – [trimesh 4.11.2 documentation](#)
10. pyproj & pyshp – Python Geospatial Libraries
- 11.11 QGIS – [QGIS Training Manual — QGIS Documentation documentation](#)
- 12.12 CesiumJS – [Tutorials | CesiumJS | Esri Developer](#)
- 13.13 JavaScript – [JavaScript Tutorial](#)
- 14.14 HTML & CSS – [Free HTML and CSS Tutorial: Learn the fundamentals in this fun, hands-on course with Per Borgen](#)

CHAPTER 10 – APPENDIX

Appendix A: Folders/Files_code of the Project

The project follows a structured directory organization to ensure modularity, clarity, and ease of maintenance. The main folders used during the implementation are summarized below:



Appendix B: Description of Key Code Modules

Main file

Acts as the central execution script that coordinates data preprocessing, solar computation, and output generation by invoking various functional modules.

Shapefile processor file

Handles the preprocessing of building footprint data, including geometry validation, coordinate transformation, and preparation of 3D-ready datasets.

Shadow solar calculator file

Implements shadow-aware solar radiation and energy calculations using ray intersection techniques and solar position modeling.

Glb rotate exporter file

Manages the conversion, rotation, and export of 3D building models into GLB format for web-based visualization.

Visualization new file

Supports integration between generated outputs and interactive visualization platforms such as CesiumJS.

Index file

Provides the web-based interface for interactive 3D visualization and user interaction.

Appendix C: Tools and Libraries Used

The following tools and libraries were utilized during the internship:

- Python
- GeoPandas, Shapely, NumPy, Pandas
- pvlib, trimesh, pyembree
- pyproj, pyshp
- QGIS
- CesiumJS
- HTML, CSS, JavaScript